PROJECT TECHNICAL SPECIFICATION

Airborne 3D RF Spatial Mapper

Autonomous Aerial Platform for 3D RF Propagation & Signal Attenuation Mapping

1. EXECUTIVE SUMMARY & PURPOSE

The **Airborne 3D RF Dead-Zone & Multipath Spatial Mapper** is an autonomous multirotor system engineered to systematically quantify, map, and analyze radio frequency (RF) propagation patterns in three-dimensional airspace. By combining an autonomous grid flight routine (powered by ArduPilot) with a dedicated payload processor, the system captures localized RF signal metrics (RSSI, SNR, and log power) synced with precise spatial telemetry (Latitude, Longitude, Altitude).

433 MHz	±0.5 m	10 Hz	3D Heatmap
LORA SPECTRUM	3D POSITION ACCURACY	DATA LOGGING RATE	PRIMARY OUTPUT

2. HIGH-LEVEL OPERATIONAL ARCHITECTURE

Ground Beacon (Transmitter)
A stationary ground station that transmits continuous, calibrated LoRa test packets at a known output power level. Serves as the constant RF emission source for spatial attenuation measurement.

Drone Diagnostic Payload
An onboard ESP32 module connected to a LoRa transceiver and MicroSD logger. Reads telemetry directly from the flight controller via MAVLink, merges signal values, and writes CSV logs.

3. SYSTEM INTERACTION WORKFLOW



4. COMPLETE SYSTEM BILL OF MATERIALS (BOM)

Subsystem	Component Description	Key Specification	Purpose
Flight Control	SpeedyBee SB-F4V3-50-STACK	STM32F405 + 50A BLHeli_S ESC	Autonomous flight & MAVLink server
Navigation	HGLRC M100 PRO GPS	u-blox M10 GNSS + QMC5883L Compass	Global 3D position & heading tracking
Airframe	F450 Quadcopter Frame	450mm Nylon Glass-Fiber + PDB	Structural core & power routing
Propulsion	2212 920KV / 1000KV Motors	1045 Dual-Blade Propellers	Lift and flight stabilization
Wiring	20 AWG Silicone Extension Wires	200°C Thermal Limit, Multi-strand	Motor-to-ESC phase extensions
Power Filtering	35V 1000µF Low-ESR Capacitor	Electrolytic, Low-ESR	Suppresses ESC inductive voltage spikes
Payload MCU	ESP32 DevKit V1	32-bit Dual-Core, SPI/UART support	Payload logic & MAVLink parser
RF Transceiver	Ai-Thinker Ra-02 (2 units)	SX1278 Chipset, 433 MHz LoRa	Beacon transmit & RSSI/SNR sampling
Data Storage	MicroSD Breakout Board	SPI Interface, FAT32 Support	High-speed CSV flight data logging
Logic Protection	4-Channel Logic Level Shifter	Bi-directional 5V to 3.3V	Protects ESP32 inputs from FC signals
Power Source	3S 11.1V 2200–3300mAh LiPo	35C Discharge Rate	Main flight power (~12-15 min flight)

5. COMPONENT SELECTION & ENGINEERING RATIONALE

SpeedyBee F405 V3 Stack
Selected for its integrated wireless Bluetooth connectivity (enabling rapid field parameter changes without laptops) and onboard MicroSD slot for primary flight diagnostic logging.

Ra-02 433 MHz LoRa
Chosen over standard 2.4 GHz Wi-Fi due to superior obstacle penetration, long-range propagation stability, and precise RSSI/SNR signal register reporting.

6. DATA LOGGING SCHEMA & CSV FORMAT

The onboard ESP32 payload merges MAVLink telemetry stream data (parsed from ArduPilot) with incoming LoRa packet reception metadata. Each entry is committed to the MicroSD card in a structured CSV format:

```
      Timestamp_ms, Latitude, Longitude, Altitude_m, RSSI_dBm, SNR_dB, Packet_ID 104820, 12.9
71598, 77.594562, 14.50, -68, 8.25, 1024 104920, 12.971601, 77.594565, 14.55, -69, 8.00, 1025
105020, 12.971605, 77.594569, 14.60, -72, 6.75, 1026 105120, 12.971608, 77.594572, 14.62, -8
5, 2.50, 1027
```

7. IMPLEMENTATION ROADMAP & EXECUTION PHASES

Phase 1: Bench Validation (Active Phase)

- Wire ESP32, Ra-02 LoRa, and MicroSD card breakout modules on breadboards.
- Flash Ground Beacon firmware to continuously broadcast 433 MHz test pings.
- Validate payload receiver RSSI/SNR computation and test dummy CSV logging on MicroSD.

Phase 2: Airframe Integration & Mechanical Assembly

- Assemble F450 frame and extend motor leads using 20 AWG silicone wiring.
- Mount SpeedyBee F405 stack with silicone grommets to isolate gyro vibration.
- Solder 35V 1000µF Low-ESR capacitor across main battery pads; wire 5V supply to ESP32 payload.

Phase 3: Autonomous Mapping & Post-Processing

- Flash ArduPilot (`SpeedyBeeF405v3`), configure u-blox M10 GPS and QMC5883L compass.
- Design 3D grid flight paths in Mission Planner and execute autonomous mapping missions.
- Export MicroSD CSV data to MATLAB / Python to render 3D spatial attenuation heatmaps.